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Sky Background Contamination Explains the Unexpectedly High Temperature of WD 1856 b

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ABSTRACT

As the only known planet transiting a white dwarf star, WD 1856 b provides unique insight into post-main-sequence planetary systems. Recent JWST/NIRSpec observations provided evidence for hydrocarbons and aerosols, alongside significant thermal emission consistent with an unexpectedly hot (390−412 K) world that may have undergone dynamical reheating (MacDonald et al. 2026). However, this temperature is in $\sim 16\sigma$ tension with previous JWST/MIRI measurements (Limbach et al. 2025). In this *Letter*, we show that the unexpectedly hot temperature is driven by uncorrected background contamination in the NIRSpec data. Using multiple methods to robustly account for this background, we recover a planetary temperature consistent with the MIRI measurements, eliminating the need for a recent reheating event. While we confirm the presence of aerosols, we estimate an upper limit on the methane abundance nearly 100 times lower than reported by MacDonald et al. (2026). The tentative evidence for a 4.3 μm phosphine feature is not recovered. Our results demonstrate that background contamination can substantially affect NIRSpec time-series observations of faint sources, suggesting that a custom reduction routine should be adopted in these cases.

1. INTRODUCTION

Though we have discovered thousands of planets outside our solar system, very little is known about the fate of these worlds as their host stars evolve off the main sequence. WD 1856+534 b (hereafter WD 1856 b) is currently the only known intact planet transiting a white dwarf. Discovered with the Transiting Exoplanet Survey Satellite (*TESS*), WD 1856 b is a giant planet orbiting just 0.02 AU from its host star (Vanderburg et al. 2020). This close-in orbit is unexpected, as the progenitor star would have engulfed the planet as it evolved unless the planet originated farther out in the system. Two principle formation pathways have been proposed to account for this: common envelope evolution (CEE, Chamandy et al. 2021; Lagos et al. 2021; Merlov et al. 2021) and high-eccentricity migration (Muñoz & Petrovich 2020; Maldonado et al. 2021; O'Connor et al. 2021; Stephan et al. 2021; O'Connor et al. 2022). Both dynamical pathways are expected to heat the planet, leaving a potentially observable signature. However, CEE is only expected to heat the planet's surface for approximately 1 Myr (Glanz & Perets 2018) following the end of the AGB phase, whereas high-eccentricity migration can occur at any point in a planet's history and may deposit energy deep in the planet's interior. Unexpected thermal excess from a planet orbiting a relatively old white dwarf, such as WD 1856 (with a cooling age of 5.4±0.7 Gyr, Limbach et al. 2025), would therefore provide evidence that the planet had undergone recent migration.

Several JWST programs have observed WD 1856 with the Mid-Infrared Instrument (MIRI) and Near Infrared Spectrograph (NIRSpec), providing constraints on the planet's physical and atmospheric properties that can help distinguish between the proposed evolutionary scenarios. MIRI imaging

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measurements, collected at a phase of $\phi \approx -0.3$, are consistent with a 184±8 K planet, which would make it the coldest planet from which we have directly detected light (GO #5204, PI M. A. Limbach, Limbach et al. 2024, 2025). Given the system's age (7−10 Gyr), this temperature implies an upper limit on the mass of $5.2^{+0.7}_{-0.8}$ M_{Jup}, confirming WD 1856 b's planetary nature. MacDonald et al. (2026) have since presented an atmospheric study of WD 1856 b using a Cycle 1 NIRSpec/PRISM time-series observation (TSO) of the planet's transit (GO #2358, PI R. MacDonald). They reported evidence for hydrocarbons, aerosols, and thermal emission from the planet's nightside. This emission is consistent with a planet temperature of 390−412 K, substantially warmer than expected given the planet's age and mass (4.3−10.9 M_{Jup}, as calculated with their atmospheric retrievals). The authors interpret this result as evidence for migration-related heating. Using dynamical models, they find that the retrieved planet temperature, size, and mass are consistent with migration occurring billions of years after the AGB phase, supporting a high-eccentricity migration scenario. However, this prediction relies on the inferred NIRSpec temperature, which is in 16−17σ disagreement with the previously published MIRI observations (with the exact σ tension depending on the MacDonald et al. (2026) data reduction pipeline used for comparison).

There are two potential ways to explain this discrepancy. First, the excess thermal emission measured with NIRSpec could be astrophysical. However, this would require invoking surprising atmospheric properties, such as a nightside temperature more than twice as hot as the dayside, 200% temperature variations on timescales of months, or extreme changes in cloud coverage across the planet. Any of these effects would be unprecedented in giant planet atmospheres. Alternatively, the tension could result from a systematic effect in either the MIRI or NIRSpec observations. A MIRI systematic would have troubling implications, as MIRI imaging is frequently used to observe similarly faint objects (e.g., Pérez-González et al. 2024; Williams et al. 2024; Östlin et al. 2025). However, MIRI observations of white dwarfs have been shown to be in good agreement with model predictions based on visible light observations (e.g., Limbach et al. 2024; Poulsen et al. 2024), demonstrating that similar MIRI observations do not exhibit this systematic. In contrast, NIRSpec TSOs of exoplanet transits have primarily targeted bright, main-sequence host stars, meaning that a NIRSpec systematic relevant for WD 1856 may not have been relevant for the more standard use cases. For programs where NIRSpec TSOs do target faint sources like substellar objects and stellar remnants, similar excess red flux has been observed (Biller et al. 2024; Luhman et al. 2024; Broski-Laing et al. 2026). If such a systematic effect existed in NIRSpec, it could significantly skew estimates of a planet's temperature because the NIRSpec bandpass covers a small fraction of WD 1856 b's bolometric flux (0.02% for a 184 K planet, compared to 40% for MIRI). This means that considerable extrapolation is needed to measure an effective temperature. Regardless of the outcome, understanding and resolving the tension between these MIRI and NIRSpec observations will be essential for both characterizing WD 1856 b and for assessing the reliability of JWST observations of faint sources.

In this *Letter*, we demonstrate that the apparent hot temperature inferred from NIRSpec/PRISM is caused by an incomplete background correction introducing contamination at red wavelengths in the NIRSpec bandpass. In §2, we describe the JWST observations of the system that could be sensitive to thermal emission from the planet. In §3, we show that the NIRSPEC/PRISM data are contaminated by significantly more background flux than estimated by standard analysis pipelines, and we describe two approaches for correcting this background and deriving uncontaminated spectra. We take the corrected spectra and perform an updated atmospheric retrieval in §4. We outline a secondary mass constraint from fitting a Saturn-like spectral feature in §5. We then discuss the implications for WD 1856 b's dynamical history in §6, alongside a discussion of how residual background signals may impact other JWST programs, before concluding in §7.

2. OBSERVATIONS

2.1. *Literature Observations*

Several JWST programs have observed the WD 1856 system, including GOs #5204, #2358, and #2507. GO #5204 imaged the system on 29 July, 2024 with seven MIRI filters: F560W, F770W, F1000W, F1130W, F1280W, F1500W, and F1800W, with integration times ranging from 1.9 to 22.6 minutes. To avoid both transit and eclipse events, observations occurred at a phase of $\phi \approx -0.3$. As a result, the dayside was slightly more visible than the nightside. Using these observations, Limbach et al. (2025) measured absolute photometry of the combined white dwarf and planet system and detected significant excess flux at 12–18 μm. These results placed the first strong constraints on the planet's temperature (184±8 K, Limbach et al. 2025).

GO #2358 observed the system on 27 April, 2023 using NIRSpec/PRISM in Bright Object Time Series (BOTS) mode for 122.8 minutes. This program captured the ∼8 minute transit event with the goal of characterizing the planet's atmosphere via transmission spectroscopy (MacDonald et al. 2026). They reduced the data using two independent data pipelines—the Fast InfraRed Exoplanet Fitting Lyghtcurve (FIREFLy, Rustamkulov et al. 2022; Rustamkulov et al. 2023; Sing et al. 2024) and a new custom pipeline called Juniper—that produced mostly consistent results. We show the Juniper spectrum in Figure 2 (available on Zenodo, MacDonald et al. 2026). The authors detected a highly significant decrease in transit depths from 3.5–5 μm wavelengths, which they attribute to dilution from the planet's thermal emission (e.g. Kipping & Tinetti 2010; Vanderburg et al. 2020). Fitting this apparent thermal emission slope yielded a temperature of 390–412 K (MacDonald et al. 2026). As previously discussed, the MIRI and NIRSpec programs report significantly different (∼16σ) planet temperatures. We compare the two datasets and corresponding planet models in Figure 1.

2.2. *Independent Observations with NIRSPEC/G395M*

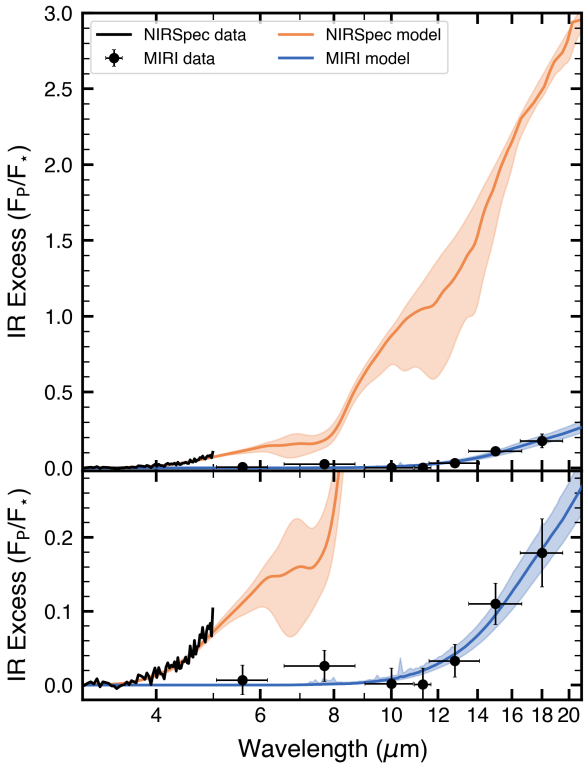


Figure 1. *Top*: NIRSpec/PRISM and MIRI imaging observations. NIRSpec data is consistent with a ~400 K planet, demonstrated by the best-fit Juniper model (orange, MacDonald et al. 2026), while MIRI data is consistent with a much colder 184 K planet (blue, Limbach et al. 2025). 1σ uncertainties are denoted by the corresponding shaded regions. The best-fit NIRSpec temperature is highly discrepant with the MIRI observations. *Bottom*: Zoomed-in comparison.

A third program that is not yet reported in the literature, GO #2507 (PI A. Vanderburg), provides an opportunity to investigate the underlying cause of this discrepancy. GO #2507 observed the WD 1856 system with the G395M grating, which measures the spectrum between 2.9–5.2 μm and provides an independent measurement of the apparent thermal slope observed by PRISM. These observations were obtained on 17 June, 2023 using 16 groups per integration and 414 total integrations, resulting in a total exposure time of 106 minutes. Observations used the SUB2048 subarray and NRSRAPID readout pattern. Like the PRISM observations, the G395M observations were taken in Bright Object Time Series (BOTS) mode. We reduce the `uncal.fits` data using the open-source `Eureka!` pipeline v 1.5 (Bell et al. 2022), a wrapper for the `jwst` pipeline that also provides additional reduction steps. We primarily use the default `Eureka!` settings, except where noted. Stage 1 performs `jwst` steps such as superbias correction, dark current subtraction, and jump detection. We adopt a jump detection threshold of 10σ. To correct for $1/f$ noise, Stage 1 also performs group-level background subtraction by removing the column-by-column median of pixels outside of the masked trace. Stage 2 then performs additional calibrations, and Stage 3 produces time-series spectra. To correct the curved trace, we identify the center of the trace in each column, assuming a Gaussian distribution, and correspondingly shift the position of each column. We use a half-width aperture of 3 pixels to extract the spectrum and use the region at least 4 pixels away from the trace to measure the background. We then apply a 10σ threshold for outlier rejection during background subtraction. In stage 4, we obtain 2D light curves. For each unbinned light curve, we remove points that deviate by more than 10σ from a rolling median.

We fit the light curves using a custom fitting routine. We first model the white light curve using the Python package `batman` (Kreidberg 2015), and fit for seven free parameters: the time of inferior conjunction T_0, planet-to-star radius ratio $R_P/R_\star$, semi-major axis $a/R_\star$, inclination $\cos i$, quadratic limb-darkening coefficients μ_1 and μ_2, and a jitter term σ. Because the tidal circularization timescale for Jupiter-mass planets is much less than the age of the system, we assume an eccentricity of zero. Additionally, we require that the planet's radius does not exceed the Roche lobe radius, thus constraining the sampler to planets that are not undergoing Roche-lobe overflow. Finally, we set Gaussian priors on $R_P/R_\star$, $a/R_\star$, and $\cos i$. The priors are centered on best-fit values from Vanderburg et al. (2020), with standard deviations corresponding to the measured uncertainties. When fitting individual, wavelength-binned light curves, we then hold all parameters fixed to the best-fit value from the white light curve fit, with the exception of $R_P/R_\star$ and σ.

We explore the parameter space with a differential evolution Markov Chain Monte Carlo (MCMC) sampler. Before calculating the likelihood at each model iteration, we subtract our transit model from the data and fit a linear trend to the residual light curve at each MCMC link. We then add this line to the transit model. This corrects for out-of-transit variations. Sampling is done using the `edmcmc` package (Vanderburg 2021). We evolve 128 chains for 2,000 iterations each, with a burn-in of 500.

In Figure 2, we compare our resulting spectrum to the published PRISM data. For an astrophysical source, we would normally expect to measure the same slope—within uncertainties—for both the NIRSpec PRISM and G395M observing modes, as they both cover the relevant wavelength regime and were executed just ~1.5 months apart. We do find that the G395M spectrum also exhibits an apparent thermal slope. However, this slope is significantly more shallow than the PRISM results. Fitting a quadratic model to both datasets within the 3–5.2 μm region (shown in Figure 2), we find a 8σ tension in the predicted transit depth at 5 μm. If this is a true astrophysical signal, then this corresponds to an unexpectedly large change in planet's nightside temperature on a short timescale. However, a second explanation is that

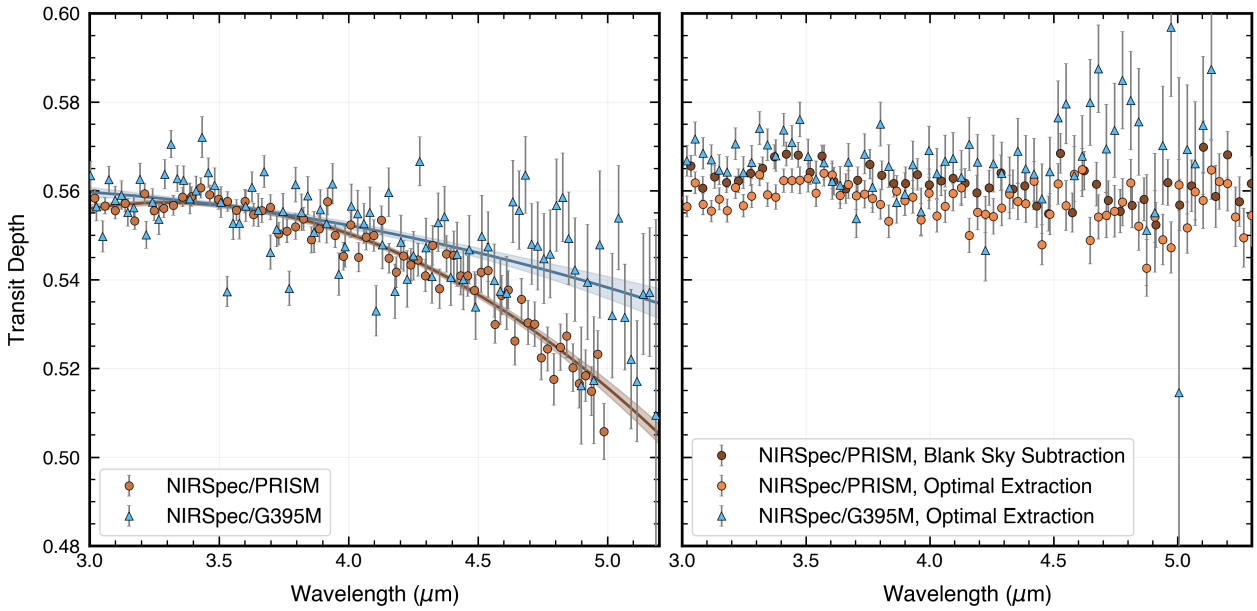


Figure 2. *Left*: Comparison between NIRSpec/PRISM and NIRSpec/G395M data, denoted by orange circles and blue triangles, respectively. To guide the eye, we also show best-fit quadratic fits to both datasets within the 3−5.2 μm region, with corresponding 1σ uncertainties denoted by the shaded regions. *Right*: Corrected NIRSpec/PRISM observations. We subtract the background emission using a blank-sky observation (light orange) and an optimal extraction pipeline (dark orange). We find that both methods largely eliminate the apparent thermal slope.

there is a systematic at these wavelengths in NIRSpec that manifests differently depending on the observing mode and reduction method, and is not being properly accounted for in either dataset. In this case, information about the thermal emission inferred from this region should be interpreted with caution.

3. BACKGROUND CORRECTION

The discrepancy between these three datasets suggests there may be an underlying issue with the NIRSpec TSO observations. In this section, we show that this discrepancy is likely due to uncorrected background light contaminating the NIRSpec results. First, we use a blank sky observation to obtain an empirical per-pixel estimate of the background. We compare this to the estimated background from MacDonald et al. (2026) (§3.1) and show that the background used in the NIRSpec analysis differs considerably from our empirical spectrum. We demonstrate that a simple sky subtraction using our empirical background from the blank sky observation can largely eliminate the apparent thermal slope (§3.2). We then describe a custom reduction pipeline that accounts for the local sky background using empirically derived trace profiles to perform optimal spectral extraction (§3.3).

3.1. *Empirical Background Estimate*

NIRSpec time series data analyses typically make a column-by-column estimate of the background flux in an observation based on the flux measured in pixels far from the trace of the star. Typical analyses take the average or median of pixels located far from the star, assume this background flux level is constant along the column, and subtract the background from all pixels in the column, including the pixels containing flux from the star. However, if the background is not constant along the column, or the pixels used to estimate the background are not representative of the full column's behavior, then this method can fail.

We therefore make an independent measurement of the background flux measured by NIRSpec/PRISM TSOs. To do this, we take advantage of a failed PRISM TSO from GO program #3496 (PI J. Vos). This observation consisted of a 15.5 hour PRISM time series, and was intended to observe a brown dwarf. However, due to a target acquisition failure, it instead observed blank sky. This observation therefore provides an opportunity to empirically measure the background flux in the pixels where starlight would normally prevent the background from being measured directly. To make this estimate, we reduce the blank sky data using the Juniper pipeline[1] and configuration files from MacDonald et al. (2026, private communication). However, we exclude the background subtraction step. The resulting empirical background spectrum is shown in Figure 3, alongside a WD 1856 stellar model spectrum (Limbach et al. 2025; MacDonald et al. 2026). For comparison, we also show the column-by-column estimate of the background subtracted during Stage 3 of the Juniper pipeline for the WD 1856 b observations, using the same background regions as MacDonald et al. (2026). We find that in most of the spectral range, the flux in the empirical

[1] https://github.com/AbbyBoehm/Juniper

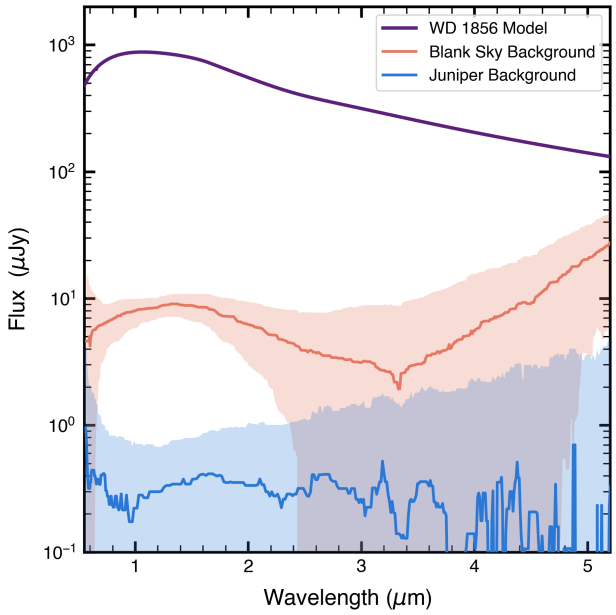


Figure 3. Comparison between the modeled WD 1856 spectrum (purple), the median empirical blank sky spectrum from GO #3496 (orange), and the median Stage 3 background estimated using the Juniper pipeline (blue). Uncertainties are denoted by corresponding shaded regions. We find that the predicted background does not match the profile of the observed sky background, suggesting there is excess flux in the ~3–5 μm region.

background spectrum is orders of magnitude smaller than the flux of WD 1856. However, at long wavelengths, contributions from the background are higher and the flux of the star is lower, meaning that the relative importance of background subtraction increases. At the reddest wavelengths, the background flux is approximately 20% of the stellar flux – enough to explain the apparent thermal slope in the NIRSpec/PRISM observations.

In contrast, the estimated background flux from the Juniper pipeline for WD 1856 b is relatively constant with wavelength, and is at least an order of magnitude lower than the flux from our empirical estimate. While the blank sky observations may differ from WD 1856's background due to differences such as sun angle and zodiacal light (see §3.2), the fact that the background profiles are discrepant suggests that a different method of background subtraction may produce significant differences in the resulting transmission spectrum. We investigate this in the following sections.

3.2. *Blank Sky Subtraction*

We provide a simple demonstration that it is possible to remove the red slope in the NIRSpec/PRISM data by subtracting this sky background. We subtract the Juniper Stage 3 blank-sky background from the Juniper Stage 3 science observation (reduced following the description in MacDonald et al. 2026), allowing the background to scale by a multiplicative factor f. This allows for non-uniform background contributions, such as variable zodiacal light. To estimate the value of f, we select regions around the trace where we expect minimal contamination from the target. To minimize light from the white dwarf, we exclusively use the four integrations that occur closest to the center of transit. We then calculate the median flux within this region for both datasets, and use the resulting ratio to estimate $f = 0.58$. Adopting this value, we subtract the sky background and perform Stage 4 of the reduction as normal. As demonstrated in §3.3, we find no strong evidence of chromatic limb darkening. We therefore fit the resulting light curves using the methods described in §2, which uses the best-fit quadratic limb darkening parameters from the white light curve. We find that this treatment eliminates the red slope, as shown in Figure 2.

This analysis demonstrates that the apparent thermal slope in the NIRSpec observations of WD 1856 b can be reproduced by background flux, and that proper removal of that flux can remove this feature from the transmission spectrum. This has important implications for the properties of the planet, which we will discuss in detail in later sections of this *Letter*. However, we note that while the analysis we have performed is illustrative, the manner in which we corrected the background introduces several complications. First, it is unclear if the blank sky observation has the same profile as sky behind WD 1856. Second, the multiplicative factor varies depending on the exact regions used to calculate it, making it difficult to identify the correct value. Ideally, we would like to find a way to separate the background component from the spectrum without the need for a reference observation. We describe a method to do this below.

3.3. *Optimal Extraction*

We implement a custom pipeline (Brandt et al. in prep.) that uses optimal spectral extraction and empirical trace profiles to robustly fit the local sky background. This approach produces ratios of spectral intensity at different epochs that are independent of the true background level. This pipeline first corrects for classic nonlinearity and the brighter-fatter effect (BFE). We use a variable number of reads per integration to calculate up-the-ramp slopes, resulting in uniform BFE (and thus uniform trace profiles) across integrations, and fit Fourier coefficients to suppress $1/f$ noise (see Appendix B in Zhang et al. 2025). Additionally, when calculating the count rate for each pixel, the JWST pipeline uses a linear combination of the individual reads, with coefficients that depend on the read and photon noise for that pixel. However, this assumes that the count rate is constant throughout the observation, which does not hold true for our deep transit observation. To correct this, we adopt uniform weights for each read. While this slightly decreases the signal-to-noise ratio, it also removes non-uniform accumulation of signal as a potential source of bias and further decreases $1/f$ noise. After correcting for these detector effects, we use likelihood-based jump detection and reject outliers more than 5σ from the rolling median.

We perform optimal spectral extraction using empirical trace profiles. To derive these profiles, we take the median

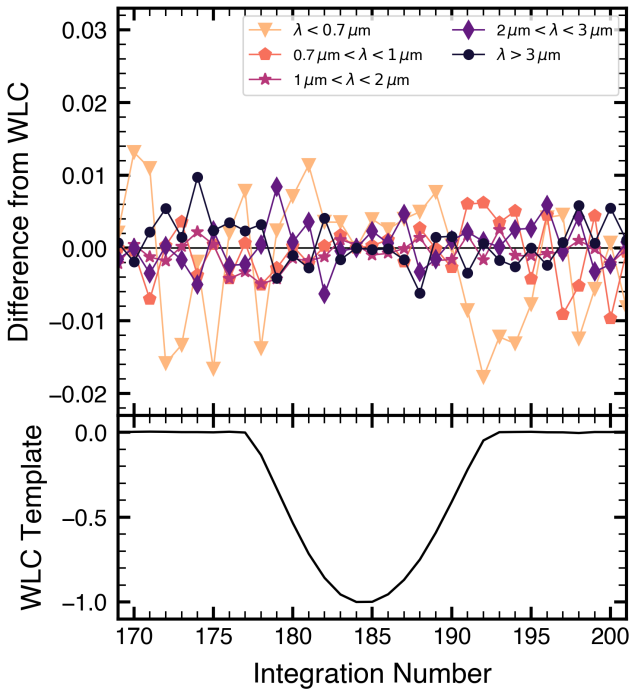


Figure 4. Chromatic limb darkening check. We compare the difference between the empirical light curves in different wavelength bins relative to the white light curve (WLC). We find that chromatic limb darkening will have a negligible effect for wavelengths > 0.7 μm, but may cause us to underestimate the transit depths at PRISM's bluest wavelengths.

of each pixel across 140 out-of-transit integrations. We select integrations centered on the transit in order to minimize systematics due to the target drifting along the slit. We use purely empirical trace profiles in the seven rows centered on row 15, the approximate center of the trace. To calculate the spectrum for each individual integration, we then fit each individual measured trace profile m in each integration as a scale factor a times the empirical profile f plus an offset b, minimizing

$$\chi^2_j = \sum_{\text{rows } i} \frac{\left(m_{i,j} - a_j f_{i,j} - b_j\right)^2}{\sigma^2_{i,j}} \quad (1)$$

where j denotes the column (i.e. wavelength) and i denotes the row. This accounts for any background that is spatially uniform across the slit. We use the best-fit scale factors a_j to produce a spectrum for each integration. Though these spectra are not in physical units, the dimensionless ratio between any two spectra at different integrations is preserved. To obtain the final transmission spectrum, we fit a scaled, normalized white-light curve template to each wavelength j. We include a first-order geometric correction to account for variations in the transit duration as a function of the planet's effective radius.

An additional potential source of chromatic variation, which we do not account for, is due to limb darkening. However, because the transit is so deep, covering approximately one hemisphere of the white dwarf, we expect limb darkening to have little impact on the observed transit depth. We can verify this in two ways: by deriving a model correction as a function of wavelength, or by comparing the empirical light curve at different wavelengths to search for evidence of chromatic limb darkening. We adopt the second approach here, computing the median empirical light curve in five different wavelength bins. We normalize each curve to a maximum depth of 100%, and subtract the unocculted level. The shortest wavelength bin is $\lambda < 0.7\,\mu$m, with a relatively poor signal-to-noise ratio. As shown in Figure 4, we find no compelling evidence of chromatic limb darkening in the empirical light curves.

At wavelengths >0.7 μm, we set an upper limit of $\sim 10^{-3}$ on the typical effect on the light curve. This contribution will be further diluted for the transit depth measurements, both because it will not be uniform over the light curve and because the peak transit depth is less than 100%. The overall effect will therefore be small relative to our uncertainties. At wavelengths <0.7 μm we expect the effect of limb darkening to be larger and the upper limits we can set are less stringent. At these wavelengths, an overall impact of as much as $\approx 10^{-3}$ on transit depth is possible. The net effect would be that a wavelength with stronger limb darkening would have a narrower light curve, resulting in positive residuals on either side of peak depth in Figure 4 and a slight underestimate of transit depth. It is therefore possible that our derived transmission spectrum slightly underestimates the observed upturn toward the bluest wavelengths.

The final spectrum, shown in Figure 2, no longer displays evidence for a downturn at around 4 μm. While the blank sky subtraction provides an important illustrative demonstration that it is possible to remove the red slope using sky subtraction, our custom reduction is explicitly built to account for contamination from the local sky background and provides direct evidence that sky background is driving the observed systematic. Additionally, this optimal extraction method avoids the two potentially troublesome assumptions made when using the blank sky observations to correct the backgrounds: that the blank sky observation has the same profile as the sky behind WD 1856, and that we have selected the optimal multiplicative factor to scale the background. We therefore consider optimal extraction a more robust approach, even though both methods produce similar background-corrected spectra. We adopt the optimal extraction results for the following analyses in §4 and §5.

Figure 2 also shows a version of the G395M spectrum reduced using the same optimal extraction methodology described here. As for the PRISM observations, the optimal extraction produces a spectrum free of the apparent thermal slope in the G395M observations. We note that our use of a fully empirical template lightcurve means that our transit depths are normalized to the largest empirical depth. This will differ slightly from the theoretical maximum depth, and results in a small, free scaling factor for our final spectra. This undetermined scaling factor is likely to be the cause of

the ≈1% overall apparent mismatch between the depths of the PRISM and G395M observations.

4. ATMOSPHERIC RETRIEVALS

Because our background-agnostic spectrum largely removes the apparent thermal slope that MacDonald et al. (2026) used to infer a high temperature for WD 1856 b, we expect that the new spectra should be consistent with a significantly colder planet. A change to the temperature profile affects the atmospheric scale height, which may in turn have important consequences for the measured abundances and planet mass. We therefore perform three sets of updated atmospheric retrievals on our custom-reduced PRISM spectrum (described in §3.3), both with and without the MIRI data, plus a separate retrieval on our G395M spectrum. The first retrieval fits only our updated NIRSpec spectrum, the second retrieval fits the combined NIRSpec and MIRI data, and the third retrieval fits the combined NIRSpec and MIRI data while masking a poorly modeled spectral region. In §4.1, we describe the setup of the retrieval frameworks used for our combined transmission spectrum and planetary thermal emission fitting. In §4.2, we discuss the findings of the retrievals and compare them to the results found in MacDonald et al. (2026).

4.1. *Methods*

We analyze the corrected NIRSpec/PRISM spectrum, calculated with our optimal extraction reduction pipeline. We use a custom Bayesian retrieval framework that acts as a wrapper around `POSEIDON`. We then call `POSEIDON` for atmospheric structure, opacities, and thermal radiative transfer (MacDonald & Madhusudhan 2017; MacDonald & Lewis 2022; MacDonald 2023) and perform our sampling using a custom wrapper code and `MultiNest`. We use the full span of data from 0.527–5.413 μm, but bin the data into 173 bins with a resolving power of $R_{\rm bin} = 75$ (similar to the resolution in MacDonald et al. 2026). Within each bin, we calculate a weighted average of the original transit depth measurements, giving more weight to measurements with smaller uncertainties, and propagate their uncertainties in quadrature (using the same weights). The binned data is shown in the top panel of Figure 5, alongside the retrieved spectrum. We use model spectra that were computed over a 0.4 – 6.0 μm spectral range at $R = 8000$. We convolve this with the wavelength-dependent Gaussian line-spread function (LSF) from Brandt et al. (in prep.), and adopt the same binning as was used for the dataset.

For our setup, we use a modeled white dwarf spectrum described by Limbach et al. (2025) and MacDonald et al. (2026), with $T_{\star,\rm eff} = 4920$ K, $\log g_\star = 8.05$, $R_\star = 0.0121\,R_\odot$, and distance $d = 24.8$ pc. The white dwarf model is based on the NIRSpec/PRISM observations, but only uses the data shortward of 3.5μm, where background contamination is not significantly impacting the data.

We use 240 layers from 10^{-7} to 100 bar to model the planet's atmosphere. We define the reference radius to be at 10 bar. When we are modeling the nightside emission, we use the same one-dimensional temperature profile, atmospheric composition, and aerosols that were used for the transmission spectrum. We include the following species in our modeling: CH_4, NH_3, H_2O, CO_2, CO, HCN, H_2S, and PH_3, with H_2 and He (at an abundance of $X_{\rm He}/X_{\rm H_2} = 0.17$). We do not include C_2H_6 (ethane) because, while we do believe there is opacity in WD 1856 b's atmosphere within the ethane band, this is coincident with the unknown hydrocarbon feature at ~3.4 μm. However, we do not believe ethane can explain this feature, as inclusion of ethane in our modeling produces unrealistically high abundances. We conclude that this feature is more logically modeled empirically by the similar features in Saturn's atmosphere (see §5). We also include Rayleigh extinction, collision induced absorption (CIA), an opaque gray cloud deck, and a power-law haze using the methods described by MacDonald & Madhusudhan (2017).

The reference P–T profile has three temperature anchors at 10^{-5}, 0.1 and 100 bar, with linear interpolation in $\log P$ and an isothermal extension above the upper anchor. We retrieve on $T_{\rm upper}$ and allow only positive temperature gradients at the higher pressures (no inversions are allowed as in MacDonald et al. 2026). Our priors are listed in table 1, and each retrieval has 17 free parameters. We broadly adopt the same priors as MacDonald et al. (2026), but we use a less complex P-T profile, as introducing additional poorly constrained increments in temperature could make the inferred temperatures and cloud-top pressures more sensitive to the chosen prior.

We also use one additional parameter not used in MacDonald et al. (2026): the maximum geometric transit depth $D_{\rm ref}$, used as a proxy for impact parameter. This is because our empirical extraction method is fundamentally different than the techniques used in MacDonald et al. (2026) and it leaves an uncertain overall normalization of the transit depth. Therefore, we allow the maximum geometric transit depth to vary along with the planetary radius, which then effectively allows us to fit for the transit impact parameter. We note that MacDonald et al. (2026) parameterize their fits using impact parameter directly, but fix its value in their retrievals. We use 400 live points and an evidence tolerance 0.5 for our nested sampling. We use `POSEIDON` v1.3.2 and adopt opacities from the `POSEIDON` database, version-1.3. For CIA below the available temperature range, we use the coefficients at the lowest tabulated temperature (which is T = 200 K for H_2–H_2 and H_2–He).

4.1.1. *Accounting for the Grazing Geometry*

Because the planet is larger than the white dwarf and the transit is grazing, the observed transit depth is not well approximated by the usual $(R_p/R_\star)^2$ expression. To calculate how much of the star is covered by the planet, we adopted the grazing-transit methodology described by MacDonald et al. (2026), but implemented it in our external wrapper rather than within `POSEIDON` because of the way our retrieval is set up. We divide each of the 240 atmospheric layers into four concentric rings and calculate how much starlight passes through each one, using the relevant atmospheric opacity and the dis-

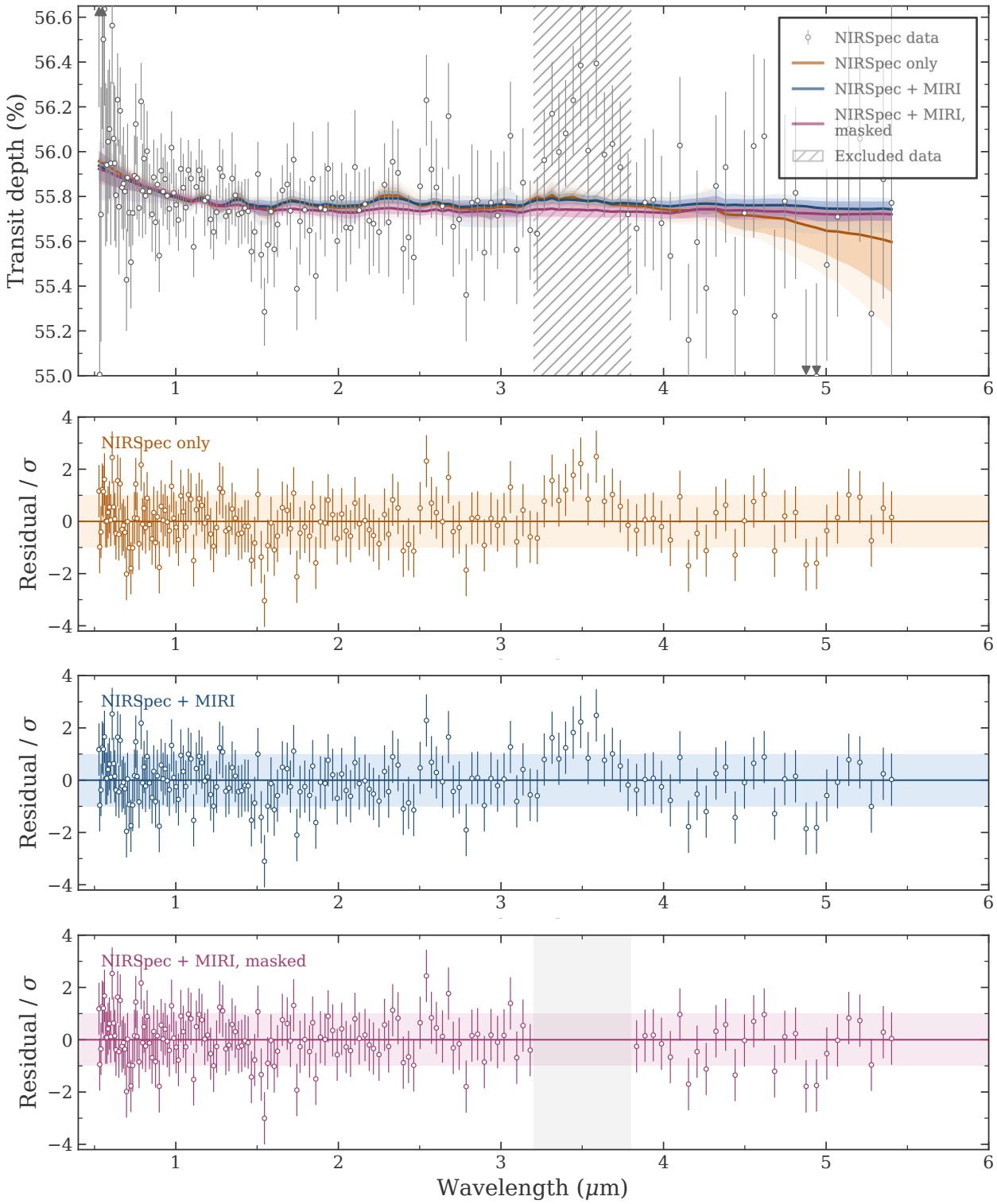


Figure 5. Measured NIRSpec/PRISM data binned to a resolution of $R_{\rm bin}$ = 75 prior to conducting the retrevial (see text for details). We present retrieved spectra from three different runs: NIRSpec only (orange), NIRSpec+MIRI (blue), and NIRSpec+MIRI excluding the 3.2-3.8 μm data (purple). We suspect that an unknown hydrocarbon feature in this spectral region, denoted by the hash marks, is artificially biasing the methane abundance to higher values. When we remove these wavelengths, methane absorption is no longer required by the data. We additionally find that the long-wavelength tail ($\gtrsim$ 4 μm) is much more strongly constrained when the MIRI observations are included in the retrievals.

Table 1. Retrieval priors

Parameter	Name	Prior
$T_{\rm upper}$ (K)	Temperature at 10^{-5} bar	$\mathcal{U}(100, 600)$
$\Delta T_{\rm mid}$ (K)	Temp change: 10^{-5}–10^{-1} bar	$\mathcal{U}(0, 500)$
$\Delta T_{\rm deep}$ (K)	Temp change: 10^{-1}–10^{2} bar	$\mathcal{U}(0, 1600)$
$R_{\rm ref}$ ($R_{\rm Jup}$)	Reference planetary radius	$\mathcal{U}(0.8451, 1.0329)$
M_p ($M_{\rm Jup}$)	Planetary mass	$\mathcal{U}(0.5, 30)$
$D_{\rm ref}$	max geometric transit depth	$\mathcal{U}(0.50, 0.60)$
$\log_{10} X_i$	Volume mixing ratio of $i^{\rm th}$ gas	$\mathcal{U}(-12, -0.3)$
$\log_{10} a$	Rayleigh enhancement factor	$\mathcal{U}(-4, 8)$
γ	Scattering slope	$\mathcal{U}(-20, 2)$
$\log_{10}(P_{\rm cl}/{\rm bar})$	Cloud top pressure	$\mathcal{U}(-6, 2)$

tance the light must travel through it. We then compute how much each ring overlaps the white dwarf. This directly provides the grazing transit depth and does not require a model for computing the planet's effective radius. The impact parameter used in our calculation is derived from the sampled reference planetary radius, $R_{\rm ref}$, and maximum geometric transit depth, $D_{\rm ref}$. More specifically, $D_{\rm ref}$ is the fraction of the white dwarf's projected disk covered at mid-transit by an opaque planetary disk of radius $R_{\rm ref}$. We allow the overall transit depth to vary during our fit, rather than fixing it from a separate measurement of the transit geometry. Thus, for computing the transmission spectrum, we use `POSEIDON` to compute the atmospheric structure/extinction in each layer (including molecular absorption, CIA, Rayleigh scattering, clouds, and haze) and then our wrapper computes the grazing solution using those layer properties from `POSEIDON` to calculate light paths, transmitted fractions, stellar overlaps, and the resulting transit depth, following MacDonald et al. (2026). To model nightside emission, we use `POSEIDON`'s thermal radiative transfer. This requires no additional changes, as the grazing geometry does not impact the emission calculation. However, the wrapper still combines the stellar and planetary fluxes to account for the planetary emission dilution, and applies the instrumental response/binning (described below).

For each unbinned wavelength channel, we calculate the blocked starlight, total starlight, and planet light separately, accounting for the instrumental PSF and spectral resolution. We then divide the blocked light by the total light from the star and planet to get the transit depth. We assume a uniform stellar intensity when calculating the overlap areas (see Brandt et al. in prep., for more discussion on the limb-darkening).

4.1.2. *Additional Retrieval Setups Beyond the Baseline Run*

In addition to our baseline NIRSpec-only setup, we conduct two additional retrievals. The first includes the planetary excess fluxes reported by Limbach et al. (2025) in the five MIRI imaging bands centered at 10.0, 11.3, 12.8, 15.0, and 18.0 μm. For this joint retrieval, we retain the 0.4–6.0 μm NIRSpec calculations described above and add a separate planetary emission calculation at $R = 2000$ over approximately 8.7–21 μm, covering the full MIRI filter responses. This results in three model calculations: the transmission over the NIRSpec wavelength range, the planetary emission over the NIRSpec wavelength range, and the planetary emission over the MIRI wavelength range. We combine the first two models to predict the observed NIRSpec transit depths and integrate the third model across the MIRI filters to predict the photometry. All three models use the same temperatures, composition, radius, and other atmospheric parameters. Thus, with the addition of the MIRI data, no additional free parameters are introduced. We add the resulting MIRI log-likelihood to the NIRSpec log-likelihood.

We note that the MIRI data were taken at a different orbital phase and include a mix of dayside and nightside emission, whereas the NIRSpec data include nightside emission only. However, it has previously been demonstrated that large (> 20 K) temperature differences between the dayside and nightside are not expected for this exoplanet (see Appendix A of Limbach et al. 2025). More importantly, we find that the temperature constraint from the NIRSpec data (when the MIRI data are not included in the fit) is driven completely by the planetary emission slope beyond 4 μm. Although our methods are greatly improved and have largely removed this background contamination, it is certainly plausible that a small sky background-induced slope could still be present, which would produce slightly overestimated temperatures for the planet. Since we know this region of the spectrum is highly susceptible to systematics and large day/night temperature differences are not physically allowed, the combined constraint from the two datasets at different phases provides our most reliable estimate of the planet's temperature with the currently available observations. Finally, considering that the NIRSpec bandpass captures very little of the planet's bolometric flux (0.02% for a 184 K planet and 9% for a 400 K planet), whereas MIRI covers a large fraction (40% for a 184 K planet and 78% for a 400 K planet), inferring a temperature from NIRSpec alone is particularly susceptible to errors because of the large amount of model SED extrapolation required. Note that the aforementioned bolometric fractions are approximated based on the model emission integrated over the wavelength ranges spanned by the observations. Specifically, for MIRI, we use the interval between the blue and red wavelength limits of the filter coverage.

Our other additional retrieval includes the same MIRI data but excludes the NIRSpec measurements from 3.2–3.8 μm. We suspect that an unidentified hydrocarbon feature is in this region (see §5), which is not represented by our adopted opacities and therefore may bias our retrieved atmospheric parameters (in particular the methane abundance since it also has a spectral feature at these wavelengths, Niraula et al. 2025). We remove the affected wavelength channels (all those in the 3.2–3.8 μm region) before binning, and calculate the binned depths and uncertainties from the remaining measurements. Apart from this data exclusion and the corresponding update

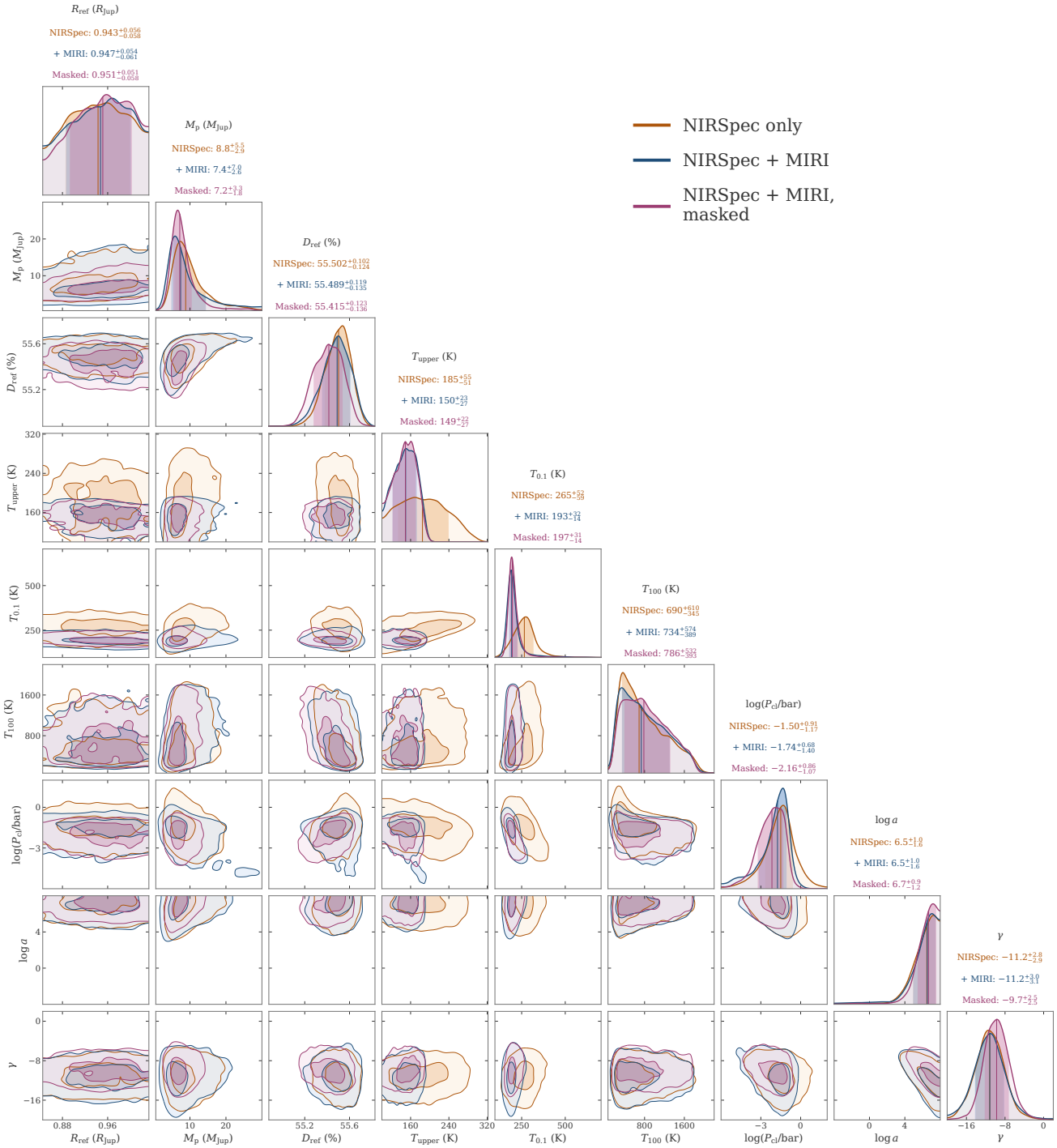


Figure 6. Derived physical parameters for WD 1856 b from the spectral retrievals using the NIRSpec-only data (orange), the combined NIRSpec+MIRI retrieval (blue), and the masked NIRSpec+MIRI retrieval (purple). We note that the addition of the MIRI data tightens the two upper-atmosphere temperature constraints.

to the binning, the atmospheric model, priors, spectral grids, and sampling settings are identical to those of the joint NIRSpec+MIRI retrieval. Because this retrieval uses the MIRI data to constrain the temperature, and also excludes a region that our model cannot effectively reproduce, we consider these results the most reliable and, unless otherwise specified, we quote them throughout the rest of the paper. The results of all retrievals are presented in Figure 5.

Finally, we also run a retrieval on the NIRSpec/G395M-only spectrum reduced with the optimal extraction technique. We bin this data to the same $R = 75$ resolution as the PRISM data, and keep the same priors and sampling as in the PRISM runs, but only compute the models over the shorter 2.8-5.3 micron range, still with models at $R = 8000$. We use a Gaussian LSF of 2.2 detector pixels (FWHM). However, this dataset is relatively noisy and the only meaningful constraint it provides is on the planet's effective temperature. Therefore, we only present the effective temperature posterior from this run (e.g., Figure 8) and omit other posteriors for clarity.

4.1.3. *Bolometric Flux and Effective Temperature*

After running our retrievals, we calculate the planet's bolometric luminosity and the resulting effective temperature using 300 randomly selected posterior samples. For each sample, we recompute the atmospheric emission model at $R = 2000$ over 0.4–50 μm. We bound the omitted thermal emission below 0.4 μm using blackbody limits. We extend the spectrum to 200 μm using a simple Planck tail matched to each sample's brightness temperature and emitting area over 45–50 μm. This is a good approximation for similarly cold planets like Saturn and Jupiter in the mid- to far-IR (Roman 2023).

From the modeled 0.4–200 μm planet emission, we estimate the planet's bolometric flux $F_{\rm bol}$ as:

$$F_{\rm bol} = \frac{L_{\rm bol}}{4\pi d^2} \tag{2}$$

and the corresponding effective temperature $T_{\rm eff}$ as:

$$T_{\rm eff} = \left(\frac{L_{\rm bol}}{4\pi R_{\rm ref}^2 \sigma_{\rm B}} \right)^{1/4}, \tag{3}$$

where $R_{\rm ref}$ is the radius associated with each sample, $L_{\rm bol}$ is the bolometric luminosity, d is the distance to the system and $\sigma_{\rm B}$ is the Stefan–Boltzmann constant. Interpreting $L_{\rm bol}$ as the global luminosity assumes homogeneous emission. We present the resulting temperature distributions in Figure 8.

4.2. *Retrieval results*

The retrieved physical parameters and P–T profile are shown in Figures 6 and 7, respectively. First, we note a few parameters that have remained consistent (within $\sim 1\sigma$) with the results from the original retrieval reported in MacDonald et al. (2026). Notably, we still find a similar aerosol slope at the bluest wavelengths. This is the portion least impacted by the background contamination, so this consistency

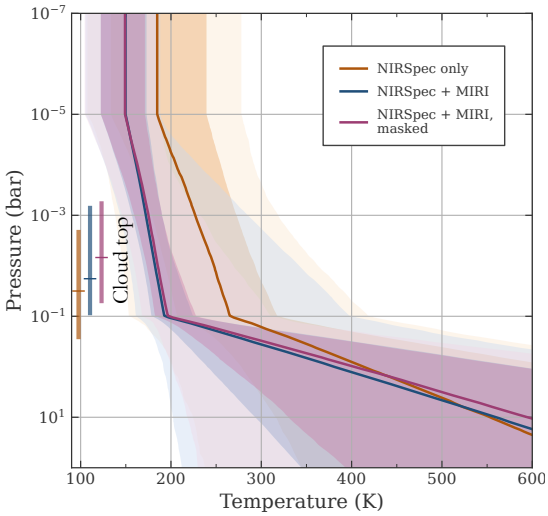


Figure 7. Retrieved P-T profiles for NIRSpec/prism-only (orange), NIRSpec+MIRI (blue), and masked NIRSpec+MIRI (purple) cases. We use a 3-node temperature profile (at 10^{-5}, 0.1 and 100 bar) for our retrievals. The three error bars on the left are the median cloud top pressure with $\pm 1\sigma$ confidence for the three cases.

is expected. We find the γ scattering parameter is consistent, and the cloud top layer is more uncertain, with the retrieved values favoring lower pressures, although there is still overlap in the posteriors. This, however, may just be a consequence of our considerably simpler P–T profile. Given that this feature empirically looks similar to that in the original reduction and is still detected robustly, we do not attempt to reconcile these small differences in aerosol parameters. Our main focus in this *Letter* is resolving the temperature discrepancy and identifying the origin of the red slope. We note that new NIRSpec/PRISM and MIRI/LRS transit observations from JWST Cycle 4 program GO #9033 (MacDonald et al. 2025) should enable more precise aerosol and chemical constraints, and determine if a more complex P-T profile is warranted by the data.

4.2.1. *Temperature and Mass*

The effective temperature of the planet changes dramatically with the new reduction, with important consequences for our understanding of the system's evolutionary history (see Figure 8). Even our NIRSpec/PRISM-only retrieval favors a much cooler temperature structure — while MacDonald et al. (2026) estimate a temperature of ~450 K at a pressure of 0.1 bar, we find $T(0.1\,\text{bar}) = 265^{+52}_{-59}$ K. The new bolometric effective temperature posterior distribution is very broad, and is consistent with MIRI results at the 2σ level (see Figure 8). Including the MIRI observations in our retrieval substantially tightens the effective temperature constraint – and completely drives the effective temperature – while leaving the other atmospheric properties largely unchanged. This is not surprising given that NIRSpec only covers ~0.02% of a cold planet's bolometric flux, whereas MIRI covers about 40%. We find that the effective temperature derived from the

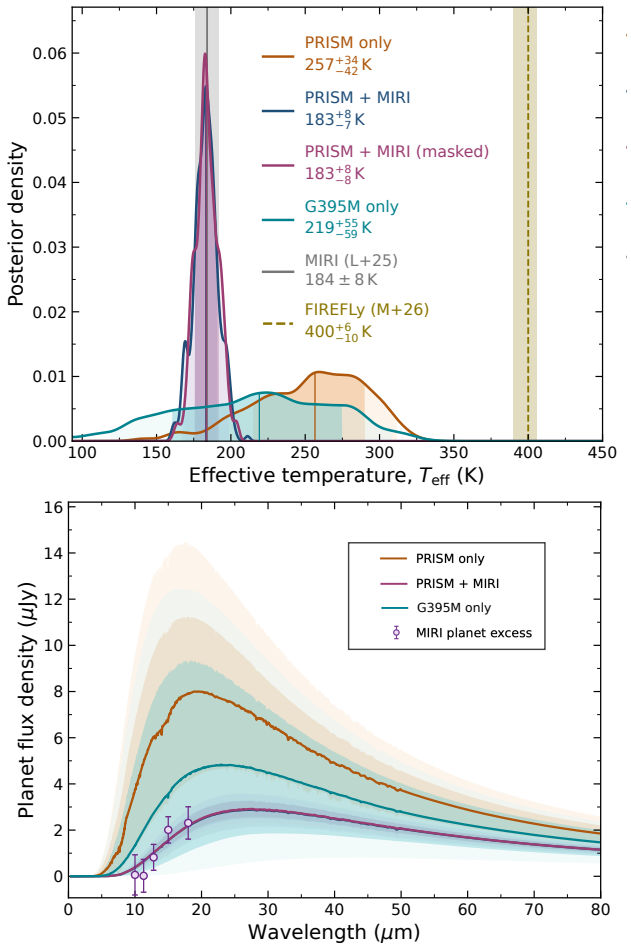


Figure 8. *Top:* Effective temperature computed from the model-integrated bolometric luminosity of our four model fits: NIRSpec/PRISM-only (orange), joint NIRSpec+MIRI (blue), masked NIRSpec+MIRI (purple), and G395M-only (teal) constraints. Also shown are previously reported values: MIRI only (gray, Limbach et al. 2025) and the FIREFLy (gold, MacDonald et al. 2026). The NIRSpec/PRISM and NIRSpec/G395M-only retrievals poorly constrains the temperature but are consistent (<2σ) with the MIRI reported literature value. The joint NIRSpec+MIRI retrieval supports temperatures consistent with the MIRI constraint. *Bottom:* Posterior median SEDs over the 0-80μm spectral range, again for the NIRSpec/PRSM-only (orange), joint NIRSpec+MIRI (purple), and NIRSpec/G395M-only (teal) retrievals. For a cold (184 K) planet, the MIRI data wavelength range covers about 40% of the bolometric flux from WD 1856 b whereas NIRSpec covers only about 0.02%, thus unsurprisingly the planet's temperature is poorly constrained without incorporating the MIRI data, and that MIRI completely drives the effective temperature constraint.

integrated planetary emission of our retrievals is essentially exactly consistent with the MIRI-only estimate of 184 ± 8 K, reported in Limbach et al. (2025). Specifically, we derive effective temperatures of 257^{+34}_{-42} K (NIRSpec/PRISM-only retrieval), 183^{+8}_{-7} K (joint MIRI+NIRSpec/PRISM retrieval),

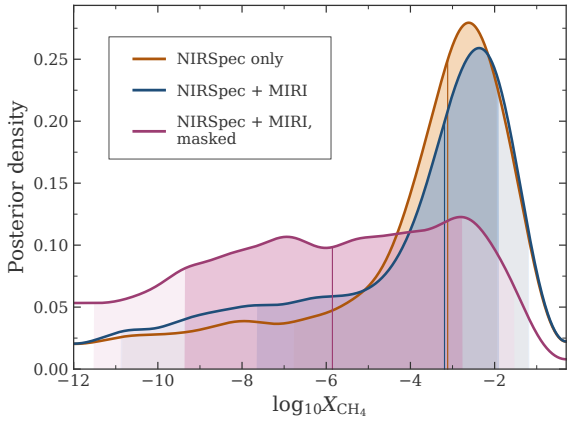


Figure 9. Methane abundance posteriors for the NIRSpec/PRISM, PRISM+MIRI and masked retrievals. If the 3.2-3.8 μm data is included, the retrieved methane abundance is $\log_{10} X_{\rm CH_4} \approx -3$, but a substantial tail to lower abundances remains. However, if the 3.2-3.8 μm data is removed, we find the methane abundance becomes an upper limit, suggesting that the detection was driven by the unknown, Saturn-like hydrocarbon feature. Methane is certainly still allowed at $\log_{10} X_{\rm CH_4} < -2.8$ (1σ upper bound), but will require additional transits to confirm the opacity and determine the abundance.

and 183±8 K (masked MIRI+NIRSpec/PRISM retrieval), and 219^{+55}_{-59} K (NIRSpec/G395M-only retrieval). There is a slight tension between the temperature from our NIRSpec/PRISM-only retrieval and our joint NIRSpec/MIRI retrievals. This may suggest that a small amount of residual background contamination could still be shifting the NIRSpec/PRISM-only retrieval toward higher temperatures. We caution against using the apparent thermal slope to infer planet temperatures in systems that are sensitive to background subtraction. We discuss the implications of this updated temperature constraint in §6.1.

Despite the colder temperature, the planetary mass remains consistent with the original analysis: $M = 7.2^{+3.3}_{-1.8}\,M_{\rm J}$ for our masked NIRSpec+MIRI retrieval, whereas MacDonald et al. (2026) report $M = 7.8^{+3.1}_{-2.7}\,M_{\rm J}$ for their FIREFLy retrieval. We note that the lower temperature may be partially offset by the lower methane abundance, which corresponds to a lower mean molecular weight. The evolutionary model mass upper limit for WD 1856 b of $5.2^{+0.7}_{-0.8}$ $M_{\rm Jup}$ (inferred from T = 184 K, Limbach et al. 2025), remains more restrictive than, but consistent with, the retrieval-based mass constraint of the planet. We also note that our retrieval has degeneracies between mass and haze properties.

4.2.2. *Methane*

Our retrievals include CH_4, NH_3, H_2O, CO_2, CO, HCN, H_2S, and PH_3. However, methane is the only species that we tentatively detect (see Figure 12). Our retrieved methane abundance is substantially lower than in the original analysis. The NIRSpec/PRISM-only retrieval gives $\log_{10} X_{\rm CH_4} = -3.12^{+1.17}_{-3.65}$, corresponding to approximately 0.08%, compared with the approximately 7% reported by

Table 2. Priors and best-fit properties from Saturn feature scaling (see §5).

Parameter	Units	Prior	Value
M_P	Mass (M_{Jup})	$\mathcal{U}(0, 20)$	4^{+2}_{-1}
R_P	Radius (R_{Jup})	$\mathcal{N}(0.95, 0.02)$	0.95 ± 0.02
T_{eff}	Effective temperature (K)	$\mathcal{N}(184, 8)$	184 ± 8
d	Optical transit depth (%)	$\mathcal{U}(0.5, 0.7)$	55.71 ± 0.06

MacDonald et al. (2026). This is nearly two orders of magnitude lower, and broadly comparable to the measured methane abundances in Jupiter and Saturn (Fletcher et al. 2009; Gapp et al. 2024). We find that including MIRI data in our retrieval has little impact on the methane abundance.

However, we find that the methane constraint does depend strongly on our inclusion of the 3.2–3.8 μm region, where our model is unable to replicate a prominent spectral feature. When that portion of the spectrum is excluded, the posterior broadens significantly toward very low abundances, and the methane detection is no longer robust (see Figure 9). Though methane remains allowed, the inferred abundance should be treated cautiously when this region is included.

4.2.3. *Phosphine*

MacDonald et al. (2026) reported tentative evidence for phosphine ($\log_{10} X_{PH_3} = -4.8^{+0.7}_{-2.1}$) near 4.3 μm, which they proposed could be an indicator of disequilibrium vertical mixing, as observed in Jupiter's upper atmosphere. However, in our re-reduction of the data we find no evidence for phosphine ($\log_{10} X_{PH_3} = -8.0^{+2.8}_{-2.5}$). We speculate that the tentative evidence in (MacDonald et al. 2026) was driven by a feature at 4.3 μm, within the region of the spectrum most affected by the background systematic. If this is true, then caution may be warranted in interpreting phosphine detections in spectra built from NIRSpec TSO mode observations.

5. SECONDARY MASS CONSTRAINT

We perform a complementary estimate of the planet's mass using the spectral feature at ~3.4 μm. Several solar system bodies exhibit a similar feature in this wavelength region, including Saturn (e.g., Dalba et al. 2015) and Titan (e.g., Robinson et al. 2014). Although the origin of this feature remains unclear, Dalba et al. (2015) suggest that an unknown hydrocarbon may be responsible, and Niraula et al. (2025) are able to model the corresponding feature in Titan's spectrum using a combination of methane, ethane, and propane. Because a planet's mass affects its scale height, it should also affect the amplitude of this feature (de Wit & Seager 2013). We therefore use the amplitude of the 3.4 μm feature to place an alternative constraint on the planet's mass. Specifically, we fit the background-corrected spectrum with a model transmission spectrum that includes thermal emission from the planet. To approximate the effect of unidentified hydrocarbons, we then add the observed ~3.4 μm spectral feature, as measured by Dalba et al. (2015).

We use the white dwarf model from Limbach et al. (2025) and (MacDonald et al. 2026), and construct the planet model from a grid of brown dwarf and giant planet emission models. These grid models are generated using `PICASO`, a 1D radiative-convective equilibrium model (Batalha et al. 2019; Mukherjee et al. 2023; Mang et al. 2026). The grid of models spans T_{eff} = 150, 250, 350, and 450 K, with log(g) = 4, [M/H] = +0.0, and C/O = 0.458. For our disequilibrium chemistry model, we use a constant strength of mixing defined by the eddy diffusion coefficient $K_{zz} = 10^7$. To model clouds, we only consider H_2O condensation and parameterize the amount of clouds forming using the sedimentation efficiency parameter $f_{sed} = 8$. We also use `PICASO` to generate the synthetic spectra. Details on the opacity tables and line list sources are detailed in Mang et al. (2026). To calculate the planet model at a given temperature, we interpolate between our emission models. We then model the planet's flux contribution using the dilution parameter described by Vanderburg et al. (2020). We manually extract the 3–4 μm Saturn feature from Dalba et al. (2015), allow it to scale it by a multiplicative factor f_S, and add it to the resulting spectrum.

We fit our corrected spectrum in the >3 μm regime, capturing both the unknown spectral feature and the planet's thermal emission. We fit for four free parameters: the planet's mass M_P, radius R_P, effective temperature T_{eff}, and optical transit depth d. At each MCMC iteration, we calculate f_S using WD 1856 b's predicted scale height H_P, Saturn's scale height H_S, WD 1856's radius $R_\star$, the Sun's radius $R_\odot$, and Saturn's radius R_S. We first approximate the amplitude of Saturn's spectral feature A_S, assuming the atmosphere is optically thick at n scale heights and $nH_S \ll R_S$:

$$A_S = \frac{(R_S + nH_S)^2}{R_\odot^2} - \frac{R_S^2}{R_\odot^2} \approx \frac{2nR_SH_S}{R_\odot^2} \tag{4}$$

We then calculate the amplitude of transmission features for WD 1856 b's specific grazing transit geometry (from Vanderburg et al. 2020):

$$A_P = \frac{2nH_P}{R_\star \pi} \tag{5}$$

Taking the ratio between these, we estimate the scaling factor:

$$f_S = \frac{A_P}{A_S} \approx \frac{H_P R_\odot^2}{H_S R_\star R_S \pi} \tag{6}$$

We then scale the Saturn feature by this value, and inject it into our planet model. We perform the sampling using the `edmcmc` package (Vanderburg 2021), and evolve 128 chains for 1,000 iterations each, with a burn-in of 250. Our best-fit model is shown in Figure 10, and all priors and best-fit parameters are provided in Table 2. We estimate a planet mass of 4^{+2}_{-1} M_{Jup}, consistent with both the original and updated constraints.

6. DISCUSSION

6.1. *Implications for the Dynamical History of WD 1856 b*

Two formation pathways have been proposed to explain WD 1856 b's close-in orbit: CEE and high-eccentricity migration.

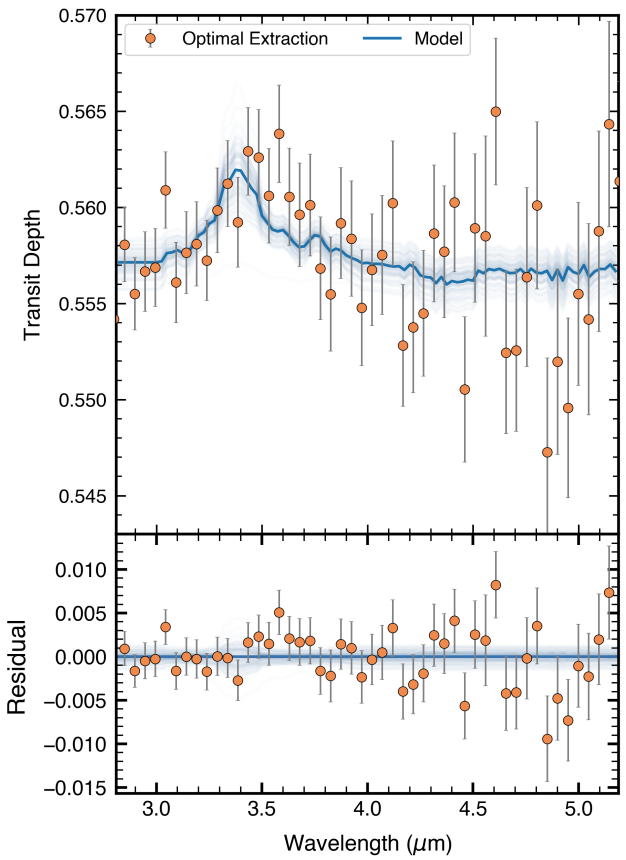


Figure 10. Best-fit model for the unidentified spectral feature at ~3.4 μm, calculated using scaled observations of a corresponding feature in Saturn's spectrum. We use this to perform a secondary measurement of the planet's mass (see §5).

A key tracer of high-eccentricity migration may be residual heat produced by the inward migration. Though CEE can also heat the planet, this signature is expected to only last for about 1 Myr (Glanz & Perets 2018) following the end of the AGB phase. MacDonald et al. (2026) therefore propose that their unexpectedly high temperature measurement may constrain WD 1856 b's dynamical history. Using cooling models for substellar objects, they find that reheating would have occurred several Gyr into the white dwarf's lifetime, ruling out CEE as an explanation. This would provide the first direct evidence that WD 1856 b has undergone high-eccentricity migration to reach its current position.

However, our NIRSpec+MIRI best-fit temperature is consistent with predictions from planetary evolution models (Fortney et al. 2007; Marley et al. 2021) for the system's age (7−10 Gyr). We therefore do not require any excess heating to explain our corrected spectrum. However, given the uncertainties in the mass of the planet and the cooling age of the white dwarf, this does not rule out a high-eccentricity migration scenario. Instead, it means that if high-eccentricity migration did occur and heat up the planet, it must have taken place far enough in the past for the planet to cool considerably.

Therefore, to reconstruct WD 1856 b's dynamical history, we will have to rely on other techniques, such as searching for atmospheric signatures of CEE (e.g., Jenkins et al. 2026) and comparing its observed properties with simulated planet populations that have undergone either CEE (Gaibor et al. *submitted*) or high-eccentricity migration (e.g., Maldonado et al. 2021).

6.2. *Relevance to Other JWST Programs*

We note that this unaccounted-for background emission is relevant for other programs targeting substellar objects and stellar remnants. In transmission-spectrum space, an additive residual background component can bias the measured transit depth in a wavelength-dependent way. To quantify this, we first calculate the in- and out-of-transit flux ($F_{\mathrm{in},\lambda}$ and $F_{\mathrm{out},\lambda}$, respectively) as a function of the true out-of-transit stellar flux $F_{\star,\lambda}$ and a residual additive background flux systematic that is not fully removed during the data reduction process S_λ. If the true transit depth is d_{true}, then the observed fluxes are given by:

$$F_{\mathrm{out},\lambda} = F_{\star,\lambda} + S_\lambda \tag{7}$$

$$F_{\mathrm{in},\lambda} = F_{\star,\lambda}(1 - d_{\mathrm{true}}) + S_\lambda. \tag{8}$$

The measured transit depth $d_{\mathrm{meas},\lambda}$ is therefore:

$$d_{\mathrm{meas},\lambda} = \frac{F_{\mathrm{out},\lambda} - F_{\mathrm{in},\lambda}}{F_{\mathrm{out},\lambda}} = \frac{d_{\mathrm{true}} F_{\star,\lambda}}{F_{\star,\lambda} + S_\lambda}. \tag{9}$$

For the limit where $|S_\lambda| \ll F_{\star,\lambda}$, this can be expanded to:

$$d_{\mathrm{meas},\lambda} \approx d_{\mathrm{true}} \left(1 - \frac{S_\lambda}{F_{\star,\lambda}}\right), \tag{10}$$

such that the observed wavelength-dependent change in the measured transit depth Δd_λ will be approximately:

$$\Delta d_\lambda \equiv d_{\mathrm{meas},\lambda} - d_{\mathrm{true}} \approx -d_{\mathrm{true}} \frac{S_\lambda}{F_{\star,\lambda}}. \tag{11}$$

This relation shows that even a relatively small background residual can generate a large apparent spectral feature in systems with very deep transits, particularly at wavelengths where the stellar flux is very low.

To estimate this effect for the WD 1856 system, we use GO #3496 to model S_λ. We take the median across all frames as the measured background systematic and use a white dwarf model for $F_{\star,\lambda}$. We then scale the background by a constant $f = 0.58$ (estimated in §3.2) to calculate the ratio between the background and stellar flux $\frac{S_\lambda}{F_{\star,\lambda}}$. We find that the contamination is negligible for wavelengths $\lesssim$3.5 μm, but reaches ~18% the flux of the white dwarf at PRISM's longest wavelengths. We now project this into transmission spectra space by multiplying by $-d_{\mathrm{true}}$, where $d_{\mathrm{true}} = 55.8\%$. The resulting spurious transmission signal is shown in Figure 11. We note that this corresponds to a completely uncorrected signal—some choices of background subtraction method may yield partial corrections, making the effect correspondingly smaller. This therefore provides an upper bound on the background's effect.

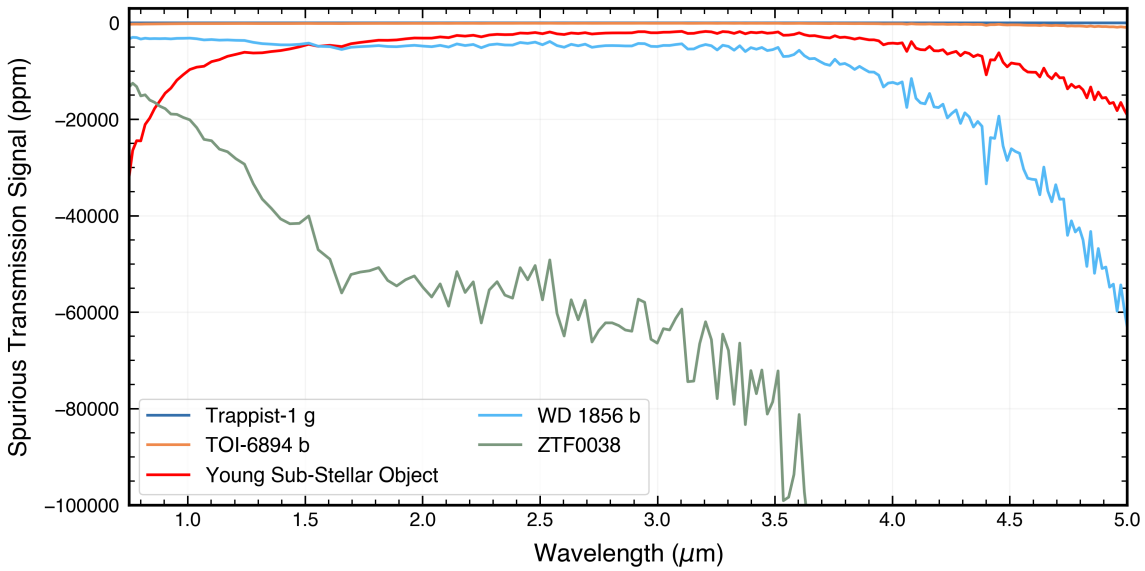


Figure 11. Spurious signal produced by background contamination Δd_λ for several systems: Trappist-1 g, a giant planet around a 0.2 $M_\odot$ star (TOI-6894 b), an eclipsing white dwarf (ZTFJ0038+2030, referred to as ZTFJ0038 in the figure), and a theoretical young, sub-stellar object with a transiting companion. We find that this contamination is most relevant for faint systems, such as white dwarfs and sub-stellar objects. We note that this assumes no background correction, and therefore provides an upper bound on the spurious signal.

Using equation 11, we can also estimate the magnitude of this contamination for other systems. We model the stellar flux as a blackbody and adopt parameters for Trappist-1 g (Agol et al. 2021), a giant planet around a 0.2 $M_\odot$ star (TOI-6894 b, Bryant et al. 2025), and an eclipsing white dwarf (ZTFJ0038+2030, van Roestel et al. 2021). We additionally model the flux from a young, sub-stellar object with a temperature of 2000 K, radius of 0.2 $R_\odot$, and distance of 300 pc, and assume a transit depth of 10%. Figure 11 shows the estimated Δd_λ for each system, calculated using the contamination method and level described here. It's clear that this signal is only a few ppm for most systems that have been observed with JWST. However, there are a few known systems where this could have an impact. In fact, other observations have similarly measured excess emission at PRISM's longest wavelengths (Biller et al. 2024; Luhman et al. 2024; Broski-Laing et al. 2026), and several upcoming or ongoing programs may also be impacted, including DDT 12507 and GOs #10300 (PI N. Whiteford), #10493 (PI K. El-Badry), #11166 (PI J. French), #11740 (PI B. Biller) and #12182 (PI S. Whitebook). For programs targeting faint stars or exceptionally deep transits, optimal spectral extraction can be used to correct this systematic (Zhang et al. 2025), and fixed slit spectroscopy may provide an alternative observational strategy for minimizing background contamination (e.g., GO #11694, PI M. Zhang).

7. CONCLUSION

We report a resolution to the 16σ disagreement in measurements of the temperature of WD 1856 b from JWST NIRSpec/PRISM (MacDonald et al. 2026) and JWST MIRI (Limbach et al. 2025). Using previously unpublished Cycle 1 JWST NIRSpec/G395M observations of WD 1856 b, we show that the apparent signature of thermal emission in the NIRSpec bandpass is not consistently reproduced by different observing modes and data reductions at the same wavelengths. Combined with the 16σ tension between the reported temperature constraints in the near- and mid-infrared, this suggests some non-astrophysical source for the observed thermal emission. In this work, we find that uncorrected background contamination in the NIRSpec/PRISM observations can replicate a planet's thermal emission. We use two independent background subtraction techniques to remove this background, and perform an updated set of three atmospheric retrievals on the resulting spectrum. These include a NIRSpec/PRISM-only fit, a joint NIRSpec+MIRI fit, and a joint NIRSpec+MIRI fit masking out the unidentified hydrocarbon feature at (3.2−3.8 μm). We also conduct a fourth retrieval on the NIRSpec/G395M-only data.

The NIRSpec/PRISM-only and NIRSpec/G395M retrievals estimate a planet temperature of 257^{+34}_{-42} and 219^{+55}_{-59} K, respectively, consistent with the MIRI results within 2σ. However, because MIRI imaging captures much of the planet's bolometric flux (40% for a 184 K planet and 78% for a 400 K planet), while NIRSpec covers only a small fraction (0.02% for a 184 K planet and 9% for a 400 K planet), we find that including MIRI observations in our retrieval substantially tightens our temperature constraint. We adopt parameters from a retrieval on NIRSpec+MIRI that masks out a region of the spectrum with a transmission feature that is not included in our models. This retrieval estimates a best-fit

temperature of 183±8 K and a corresponding mass of $7.2^{+3.3}_{-1.8}$ M_{Jup}. Additionally, the NIRSpec/PRISM-only retrieval gives a median methane abundance of 0.08‰, significantly lower than the previously reported 7‰. When we mask out the region where our model is incomplete, methane is still allowed at $\log_{10} X_{CH_4} < -2.8$ (1σ upper bound), but is no longer robustly detected. Though tentative evidence for phosphine was found at 4.3 μm by (MacDonald et al. 2026), none of our retrievals recover this feature.

Given the revised, cooler temperature, we find no evidence for dynamical reheating of the planet. However, this does not rule out a high-eccentricity migration scenario; rather, it requires that this migration occurred early enough in the white dwarf's lifetime for the planet to cool to its current temperature. We note that there are other JWST programs for which this background contamination may be relevant, including NIRSpec TSOs targeting white dwarfs and sub-stellar objects. Adopting custom background subtraction methods, such as those described in this work and Brandt et al. (in prep.), may be essential for interpreting these data.

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This research has made use of the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program.

Facilities: *JWST*.

Software: `MULTINEST` (Feroz et al. 2009), `astro.py` (Astropy Collaboration et al. 2013, 2018, 2022), `numpy.py` (Harris et al. 2020), `POSEIDON` (MacDonald 2023), `corner.py` (Foreman-Mackey 2016)

DATA AVAILABILITY

CODE AVAILABILITY

APPENDIX

A. ADDITIONAL RETRIEVAL RESULTS

Figure 12 shows the corner plot results for the chemistry of the NIRSpec-only, NIRSpec+MIRI, and masked NIRSpec+MIRI retrievals.

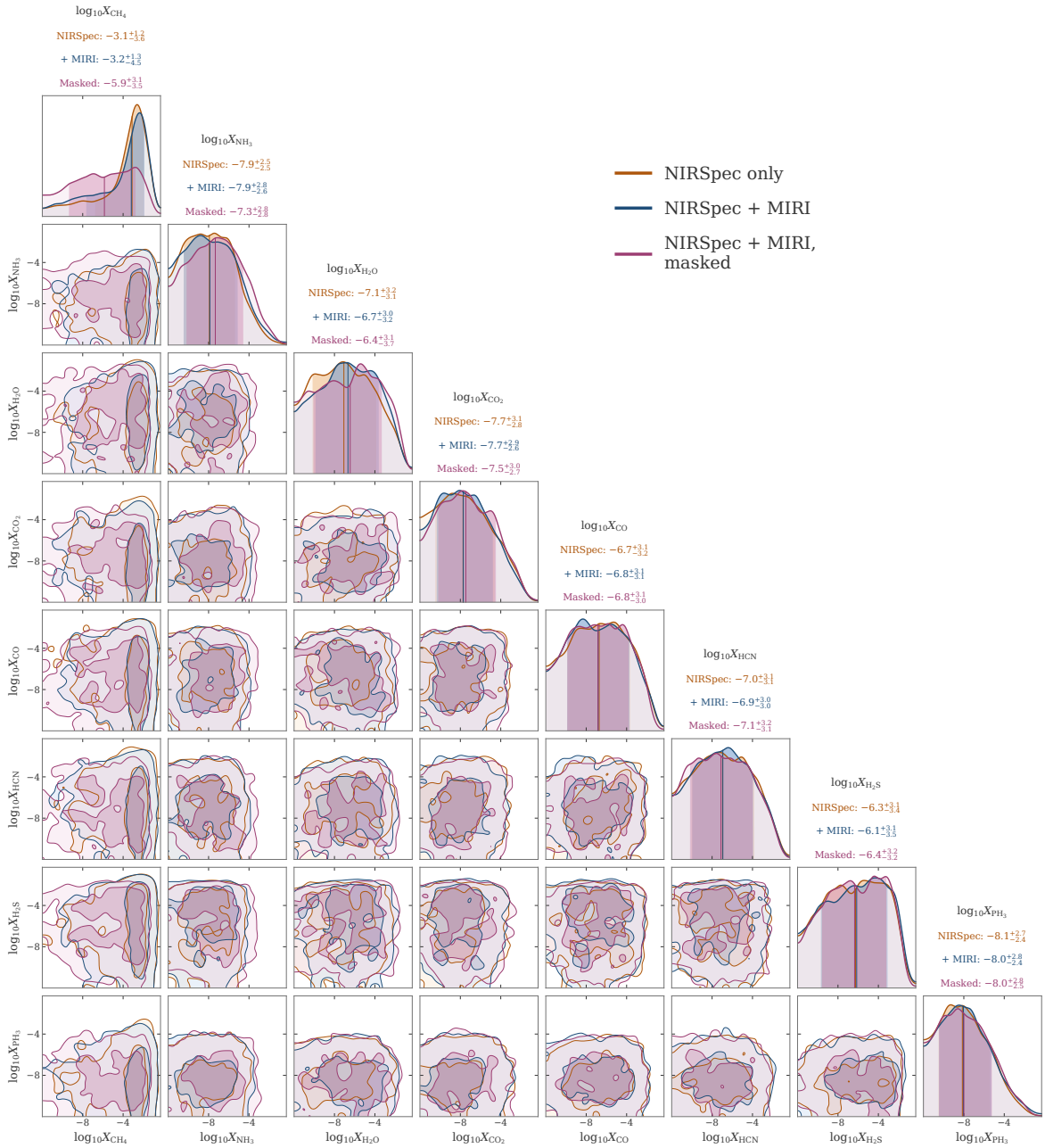


Figure 12. Chemical abundance posterior for the NIRSpec/PRISM, PRISM+MIRI and masked retrievals. The corner plot shows tentative evidence for methane opacity in the NIRSpec-only and NIRSpec+MIRI. Other than the tentative evidence for methane, no other species included in our retrieval are detected, including phosphine, which was reported with tentative evidence in MacDonald et al. (2026).

REFERENCES

Agol, E., Dorn, C., Grimm, S. L., et al. 2021, PSJ, 2, 1, doi: 10.3847/PSJ/abd022

Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33, doi: 10.1051/0004-6361/201322068

Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, AJ, 156, 123, doi: 10.3847/1538-3881/aabc4f

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Batalha, N. E., Marley, M. S., Lewis, N. K., & Fortney, J. J. 2019, ApJ, 878, 70, doi: 10.3847/1538-4357/ab1b51

Bell, T. J., Ahrer, E.-M., Brande, J., et al. 2022, Journal of Open Source Software, 7, 4503, doi: 10.21105/joss.04503

Biller, B., Chen, X., Crossfield, I., et al. 2023, Clouds or Chemistry?: Pinpointing the drivers of variability across the L/T transition via the benchmark L/T binary WISE 1049AB, JWST Proposal. Cycle 2, ID. #2965

Biller, B. A., Vos, J. M., Zhou, Y., et al. 2024, MNRAS, 532, 2207, doi: 10.1093/mnras/stae1602

Brandt, T. D., Limbach, M. A., & et al. in prep.

Broski-Laing, D., Zhou, Y., Lothringer, J. D., et al. 2026, arXiv e-prints, arXiv:2606.30112, doi: 10.48550/arXiv.2606.30112

Bryant, E. M., Jordán, A., Hartman, J. D., et al. 2025, Nature Astronomy, 9, 1031, doi: 10.1038/s41550-025-02552-4

Chamandy, L., Blackman, E. G., Nordhaus, J., & Wilson, E. 2021, MNRAS, 502, L110, doi: 10.1093/mnrasl/slab017

Dalba, P. A., Muirhead, P. S., Fortney, J. J., et al. 2015, ApJ, 814, 154, doi: 10.1088/0004-637X/814/2/154

de Wit, J., & Seager, S. 2013, Science, 342, 1473, doi: 10.1126/science.1245450

Feroz, F., Hobson, M. P., & Bridges, M. 2009, MNRAS, 398, 1601, doi: 10.1111/j.1365-2966.2009.14548.x

Fletcher, L. N., Orton, G. S., Teanby, N. A., & Irwin, P. G. J. 2009, Icarus, 202, 543, doi: 10.1016/j.icarus.2009.03.023

Foreman-Mackey, D. 2016, The Journal of Open Source Software, 1, 24, doi: 10.21105/joss.00024

Fortney, J. J., Marley, M. S., & Barnes, J. W. 2007, ApJ, 659, 1661, doi: 10.1086/512120

Gapp, C., Rengel, M., Hartogh, P., et al. 2024, A&A, 688, A10, doi: 10.1051/0004-6361/202347345

Glanz, H., & Perets, H. B. 2018, MNRAS, 478, L12, doi: 10.1093/mnrasl/sly065

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Jenkins, S., Vanderburg, A., Mang, J., et al. 2026, ApJ, 1006, 241, doi: 10.3847/1538-4357/ae83b5

Kipping, D. M., & Tinetti, G. 2010, MNRAS, 407, 2589, doi: 10.1111/j.1365-2966.2010.17094.x

Kreidberg, L. 2015, PASP, 127, 1161, doi: 10.1086/683602

Lagos, F., Schreiber, M. R., Zorotovic, M., et al. 2021, MNRAS, 501, 676, doi: 10.1093/mnras/staa3703

Limbach, M. A., MacDonald, R., Vanderburg, A., et al. 2024, Probing the Dynamical History and the Mid-IR SED of WD 1856b, JWST Proposal. Cycle 3, ID. #5204

Limbach, M. A., Vanderburg, A., Venner, A., et al. 2024, The MIRI Exoplanets Orbiting White Dwarfs (MEOW) Survey: Mid-Infrared Excess Reveals a Giant Planet Candidate around a Nearby White Dwarf. https://arxiv.org/abs/2408.16813

Limbach, M. A., Vanderburg, A., MacDonald, R. J., et al. 2025, ApJL, 984, L28, doi: 10.3847/2041-8213/adc9ad

Luhman, K. L., Tremblin, P., Alves de Oliveira, C., et al. 2024, AJ, 167, 5, doi: 10.3847/1538-3881/ad0b72

MacDonald, R., Batalha, N., Blouin, S., et al. 2025, By the Ashes of Stars: A Chemical Census of a White Dwarf Planet, JWST Proposal. Cycle 4, ID. #9033

MacDonald, R. J. 2023, The Journal of Open Source Software, 8, 4873, doi: 10.21105/joss.04873

MacDonald, R. J., & Lewis, N. K. 2022, ApJ, 929, 20, doi: 10.3847/1538-4357/ac47fe

MacDonald, R. J., & Madhusudhan, N. 2017, MNRAS, 469, 1979, doi: 10.1093/mnras/stx804

MacDonald, R. J., O'Connor, C. E., Boehm, V. A., et al. 2026, arXiv e-prints, arXiv:2607.01316, doi: 10.48550/arXiv.2607.01316

MacDonald, R. J., O'Connor, C. E., Boehm, V. A., et al. 2026, Aerosols and hydrocarbons in the atmosphere of a white dwarf planet — Additional Materials: v1.0.0, v1.0.0, Zenodo, doi: 10.5281/zenodo.18200585

Maldonado, R. F., Villaver, E., Mustill, A. J., Chávez, M., & Bertone, E. 2021, MNRAS, 501, L43, doi: 10.1093/mnrasl/slaa193

Mang, J., Batalha, N. E., Morley, C. V., et al. 2026, ApJ, 1000, 98, doi: 10.3847/1538-4357/ae47ff

Marley, M. S., Saumon, D., Visscher, C., et al. 2021, ApJ, 920, 85, doi: 10.3847/1538-4357/ac141d

McCarthy, A. M., Vos, J. M., Muirhead, P. S., et al. 2025, ApJL, 981, L22, doi: 10.3847/2041-8213/ad9eaf

Merlov, A., Bear, E., & Soker, N. 2021, ApJL, 915, L34, doi: 10.3847/2041-8213/ac0f7d

Miles, B. E., Mang, J., Morley, C. V., et al. 2026, arXiv e-prints, arXiv:2609.20664, doi: 10.48550/arXiv.2609.20664

Muñoz, D. J., & Petrovich, C. 2020, ApJL, 904, L3, doi: 10.3847/2041-8213/abc564

Mukherjee, S., Batalha, N. E., Fortney, J. J., & Marley, M. S. 2023, ApJ, 942, 71, doi: 10.3847/1538-4357/ac9f48

Niraula, P., de Wit, J., Hargreaves, R. J., Gordon, I. E., & Sousa-Silva, C. 2025, ApJL, 995, L66, doi: 10.3847/2041-8213/ae27c2

O'Connor, C. E., Liu, B., & Lai, D. 2021, MNRAS, 501, 507, doi: 10.1093/mnras/staa3723

O'Connor, C. E., Teyssandier, J., & Lai, D. 2022, MNRAS, 513, 4178, doi: 10.1093/mnras/stac1189

Östlin, G., Pérez-González, P. G., Melinder, J., et al. 2025, A&A, 696, A57, doi: 10.1051/0004-6361/202451723

Pérez-González, P. G., Barro, G., Rieke, G. H., et al. 2024, ApJ, 968, 4, doi: 10.3847/1538-4357/ad38bb

Poulsen, S., Debes, J., Cracraft, M., et al. 2024, AJ, 167, 257, doi: 10.3847/1538-3881/ad374c

Robinson, T. D., Maltagliati, L., Marley, M. S., & Fortney, J. J. 2014, Proceedings of the National Academy of Science, 111, 9042, doi: 10.1073/pnas.1403473111

Roman, M. T. 2023, Remote Sensing, 15, 1811, doi: 10.3390/rs15071811

Rustamkulov, Z., Sing, D. K., Liu, R., & Wang, A. 2022, The Astrophysical Journal Letters, 928, L7, doi: 10.3847/2041-8213/ac5b6f

Rustamkulov, Z., Sing, D. K., Mukherjee, S., et al. 2023, Nature, 614, 659, doi: 10.1038/s41586-022-05677-y

Sing, D. K., Rustamkulov, Z., Thorngren, D. P., et al. 2024, Nature, 630, 831, doi: 10.1038/s41586-024-07395-z

Skemer, A., Miles, B. E., Morley, C., et al. 2021, Water Ice Clouds and Weather on the Coldest Brown Dwarf, JWST Proposal. Cycle 1, ID. #2327

Stephan, A. P., Naoz, S., & Gaudi, B. S. 2021, ApJ, 922, 4, doi: 10.3847/1538-4357/ac22a9

van Roestel, J., Kupfer, T., Bell, K. J., et al. 2021, ApJL, 919, L26, doi: 10.3847/2041-8213/ac22b7

Vanderburg, A. 2021, avanderburg/edmcmc: v1.0.0, v1.0.0, Zenodo, doi: 10.5281/zenodo.5599854

Vanderburg, A., Rappaport, S. A., Xu, S., et al. 2020, Nature, 585, 363, doi: 10.1038/s41586-020-2713-y

Vos, J., Allers, K., Biller, B., et al. 2023, How Deep are the Winds in an Extrasolar World?, JWST Proposal. Cycle 2, ID. #3496

Williams, C. C., Alberts, S., Ji, Z., et al. 2024, ApJ, 968, 34, doi: 10.3847/1538-4357/ad3f17

Xu, S., Rogers, L., Bonsor, A., et al. 2026, A Panoramic View of Rapid Dust Production by Disintegrating Asteroids around a White Dwarf, JWST Proposal. Cycle 4, DDT, ID. #12507

Zhang, M., Bean, J. L., Coy, B. P., et al. 2024, Observing the Atmosphere of a Pulsar Planet, JWST Proposal. Cycle 3, ID. #5263

Zhang, M., Beleznay, M., Brandt, T. D., et al. 2025, ApJL, 995, L64, doi: 10.3847/2041-8213/ae157c

Zhou, Y., Amaro, R. C., Apai, D., et al. 2024, Dwarfs of Fire and Ice: Mapping the Irradiated Atmospheres of White Dwarf-Brown Dwarf Binaries with NIRSpec PRISM Phase Curves, JWST Proposal. Cycle 3, ID. #4967